1 **(a)** Correct elements in all four regions: 2 in the intersection, $4, 6, 8, 10$ in A only, $3, 5, 7$ in B only, and $1, 9$ outside both circles **[B2]**

If 0 scored, allow [B1] for 2 or 3 correct regions

(b) $\{2\}$ or 2 **[B1]**

(c) 2 **[B1]**

[Total Marks: 4]

2 **(a)** 2, 3, 5, 7 **[B1]**

(b) 1, 2, 3, 4, 6, 12 **[B1]**

(c) 8 **[B1]**

(d) 2, 3 **[B1]**

(e) 8 or 9 or 10 **[B1]**

[Total Marks: 5]

3 **(a)** $\{3, 5, 7\}$ **[B1]**

(b) 4 **[B2]**

M1 for correctly identifying $(A \cup B)' = \{4, 6, 8, 10\}$ or $A \cup B = \{1, 2, 3, 5, 7, 9\}$.

(c) $\in$ **[B1]**

(d) $\subset$ or $\subseteq$ **[B1]**

[Total Marks: 5]

4 **(a)** Intersection: $3, 5, 7$ **[B1]**

Remainder of diagram correct: A only region has 2; B only region has $1, 9$; Outer region has $4, 6, 8, 10$ **[B1]**

(b) 2 **[B1]**

(c) 6 **[B1]**

(d) $A \cap B'$ **[B1]**

[Total Marks: 5]

5 **(a)** $1, 2, 3, 6, 9$ **[B1]**

(b) $2, 3$ **[B1]**

(c) 3 **[B2]**

Partial working: Correctly identifies elements in the union $\{1, 2, 3, 5, 6, 7, 9\}$ OR elements outside $\{4, 8, 10\}$ **[M1]**

(d) $\notin$ **[B1]**

[Total Marks: 5]

6 **(a)** $2, 3, 5, 7$ **[B1]**

(b) Listing elements of $A \cup B$ as $1, 2, 3, 5, 7, 10$ **[M1]**

6 **[A1]**

(c) **(i)** $\notin$ **[B1]**

(ii) $\subseteq$ **[B1]**

[Total Marks: 5]

7 **(a)** 4 **[B2]**

M1 for $6 + 3 + 7 + 1 + 2 + 4 + 8 + x = 35$ or $31 + x = 35$.

(b) 9 **[B1]**

(c) $A' \cap B \cap C$ or $B \cap C \cap A'$ **[B1]**

(d) 16 **[B2]**

M1 for correctly identifying $n(B \cap C) = 6$ or summing $12 + 4$.

[Total Marks: 6]

8 **(a)** 7 **[B3]**

Partial working: Correct simplified equation $7x + 3 = 52$ **[M2]**

Partial working: Sets up unsimplified sum $(2x + 1) + x + (3x - 2) + (x + 4) = 52$ **[M1]**

(b) 33 **[B2]**

Partial working: Evaluates correct sub-regions for $x = 7$ (e.g. $15 + 7 + 11$) OR identifies correct algebraic expression $4x + 5$ **[M1]**

(c) Region strictly inside A but outside B is shaded (the left crescent) **[B1]**

[Total Marks: 6]

9 **(a)** Identifies correct boundary values -2 and 3 **[M1]**

$-2 < x < 3$ **[A1]**

(b) Identifying $R' = \{x : x \leq 1\}$ leading to intersection $-2 < x \leq 1$ **[M1]**

$-1, 0, 1$ **[A1]**

(c) Recognising that $P \cap S = P$ implies $P \subseteq S$ **[M1]**

$k \geq 5$ **[A1]**

(d) Identifies one correct distinct interval, e.g., $-4 \leq x \leq -1$ or $2 \leq x < 3$ **[M1]**

$-4 \leq x \leq -1$ and $2 \leq x < 3$ (accept 'or' between them) **[A1]**

[Total Marks: 8]

10 **(a)** Intersection: x and Outer region: 4 **[B1]**

P only region: $28 - x$ AND C only region: $22 - x$ **[B1]**

(b) $(28 - x) + x + (22 - x) + 4 = 45$ oe **[M1]**

$x = 9$ **[A1]**

(c) 19 **[B1]**

[Total Marks: 5]

11 **(a)** Summing all regions: $(x + 5) + 2x + x + 3 + 4 + 2 + 5 + (x + 1) = 45$ (or equivalent) **[M1]**

$5x + 20 = 45 \implies 5x = 25$ **[M1]**

$x = 5$ **[A1]**

(b) 34 **[B1]**

(c) Shading exactly the region representing P only (the top part of circle P, excluding the intersections with Q and R) **[B1]**

[Total Marks: 5]

12 **(a)** 5 in outer region **[B1]**

8 in intersection **[B1]**

17 in F only AND 10 in S only **[B1]**

(b) 17 **[B1]**

(c) $\frac{27}{40}$ **[B2]**

M1 for $17 + 10$ or 27 seen **[M1]**

(d) $\frac{8}{25} \times \frac{7}{24}$ **[M1]**

$\frac{7}{75}$ **[A1]**

[Total Marks: 7]

13 **(a)** $(x + 2) + 2 + 3 + (x - 1) + 4 + 2x + x + 1 = 36$ **[M1]**

$5x + 11 = 36$ or $5x = 25$ **[M1]**

5 **[A1]**

(b) Substituting $x = 5$ into $(x + 2) + 2 + x + 3$ **[M1]**

17 **[A1]**

(c) Identifying the correct regions: $(2x) + 3$ **[M1]**

13 **[A1]**

(d) The region strictly inside F, but outside both T and S shaded. (The 'Football only' region) **[B1]**

[Total Marks: 8]

14 (a) Intersection of all three: 8 AND Outer region: 3 **[B1]**

Two-set only intersections: 12, 7, and 10 in correct regions **[B1]**

Single set regions: x (in C), $2x$ (in D), and x (in B) **[B1]**

(b) $x + 2x + x + 8 + 12 + 7 + 10 + 3 = 100$ oe **[M1]**

$x = 15$ **[A1]**

(c) Identifies the regions required: Only C + Only D + ($C \cap D$ only) OR uses $x + 2x + 12$ **[M1]**

57 **[A1]**

[Total Marks: 7]

15 (a) 4 in center intersection **[B1]**

8, 5, 7 in the two-sport intersections **[B1]**

13, 9, 9 in the one-sport regions AND 5 in outer region **[B1]**

(b) 5 **[B1]**

(c) 17 **[B1]**

(d) $B \cap T' \cap S'$ or $B \cap (T \cup S)'$ **[B2]**

M1 for $B \cap \ldots$ **[M1]**

[Total Marks: 7]

16 (a) $2x$ in the T only region **[B1]**

x in the intersection and $3x$ in the C only region **[B1]**

2 in the region outside both circles **[B1]**

(b) 8 **[B2]**

M1 for $2x + x + 3x + 2 = 50$ or $6x = 48$.

(c) 16 **[B1]**

(d) Region outside T shaded, as well as the entirety of C (the only unshaded region is 'T only') **[B2]**

B1 for shading T' correctly OR shading C correctly.

[Total Marks: 8]

17 (a) Places 15 in the central intersection **[B1]**

Places 15 in $B \cap T$ only, 13 in $T \cap C$ only, and 7 in $B \cap C$ only **[B1]**

Places x, $2x$, $x+5$, and 8 in the respective remaining regions (B only, T only, C only, and outside the sets) **[B1]**

(b) $x + 2x + (x+5) + 15 + 15 + 13 + 7 + 8 = 103 \implies 4x + 63 = 103$ (or equivalent valid method) **[M1]**

$x = 10$ **[A1]**

(c) 85 **[B1]**

(d) $\dfrac{13}{63}$ **[B2]**

If 0 scored, allow [M1] for correctly identifying the denominator $n(T) = 63$ or the numerator $n(T \cap C \cap B') = 13$

[Total Marks: 8]

18 (a) All 8 regions correctly populated **[B3]**

Partial working: 5 to 7 regions correctly populated **[B2]**

Partial working: 3 to 4 regions correctly populated (e.g. places $12, 15, x$ correctly) **[B1]**

(Correct values are: Center = 12. $M \cap D$ only = 13. $D \cap A$ only = 20. $M \cap A$ only = 16. M only = 9. D only = 20. A only = x. Outside = 15)

(b) 15 **[B2]**

Partial working: Forms correct sum $9 + 20 + x + 13 + 16 + 20 + 12 + 15 = 120$ OR $105 + x = 120$ **[M1]**

(c) 70 **[B2]**

Partial working: Identifies the 20 students in only Drama and Art OR correctly calculates $50 + 20$ **[M1]**

(d) Region strictly inside M but outside both D and A is shaded (the top crescent / "only M" region) **[B1]**

[Total Marks: 8]